

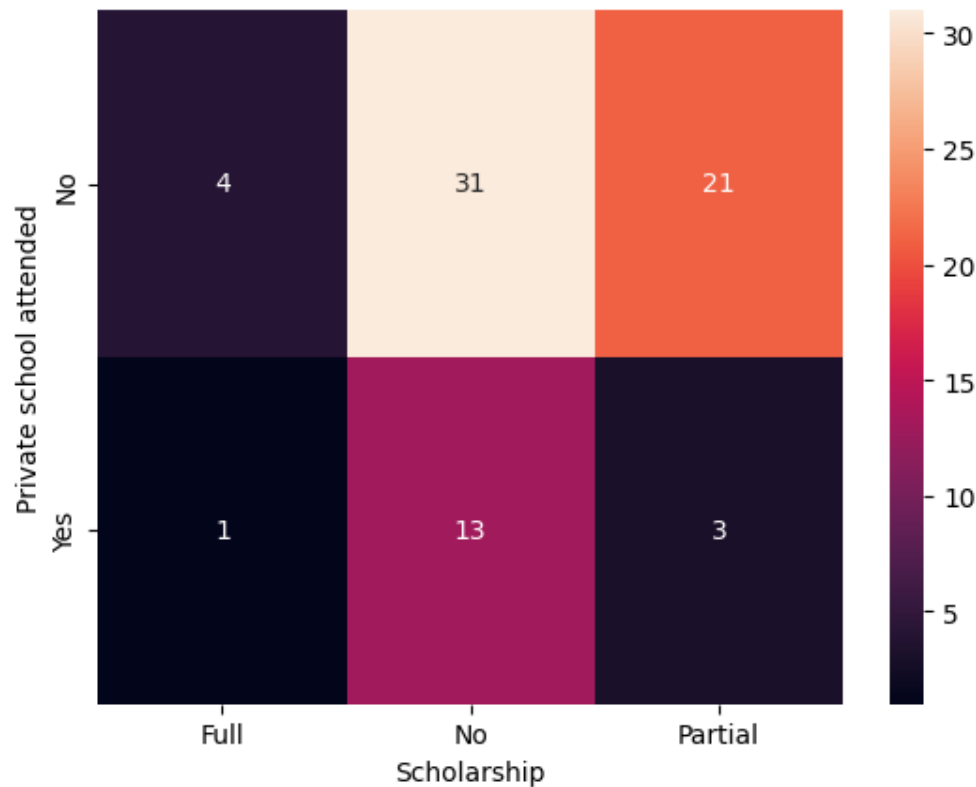


Factors affecting college students

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The dataset

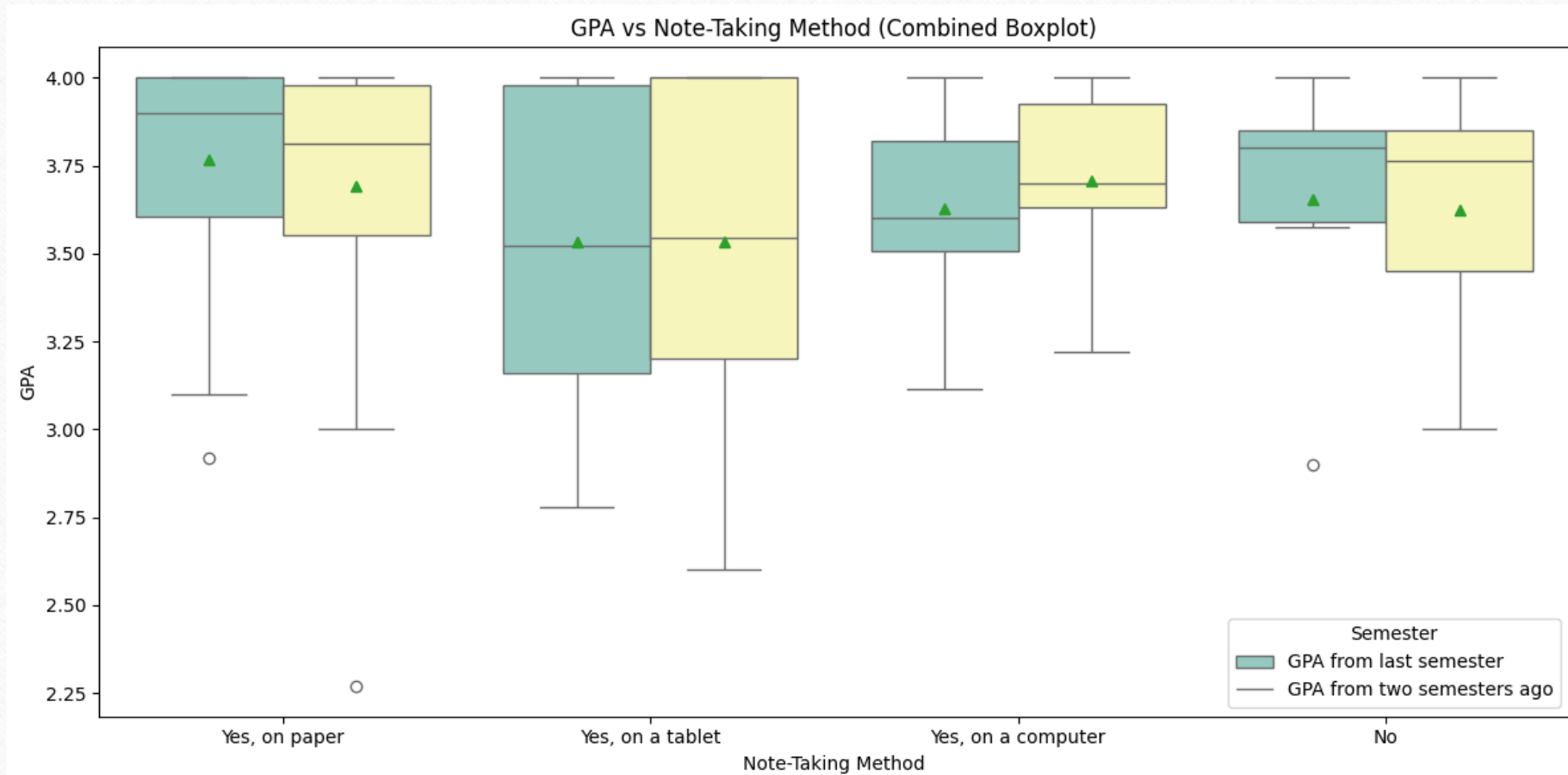
- For the observations made in this presentation, a dataset from a survey of 73 UMD students has been used.
- Considering the loss of data due to missing information, a range of 68-72 students' responses has been used per hypothesis.



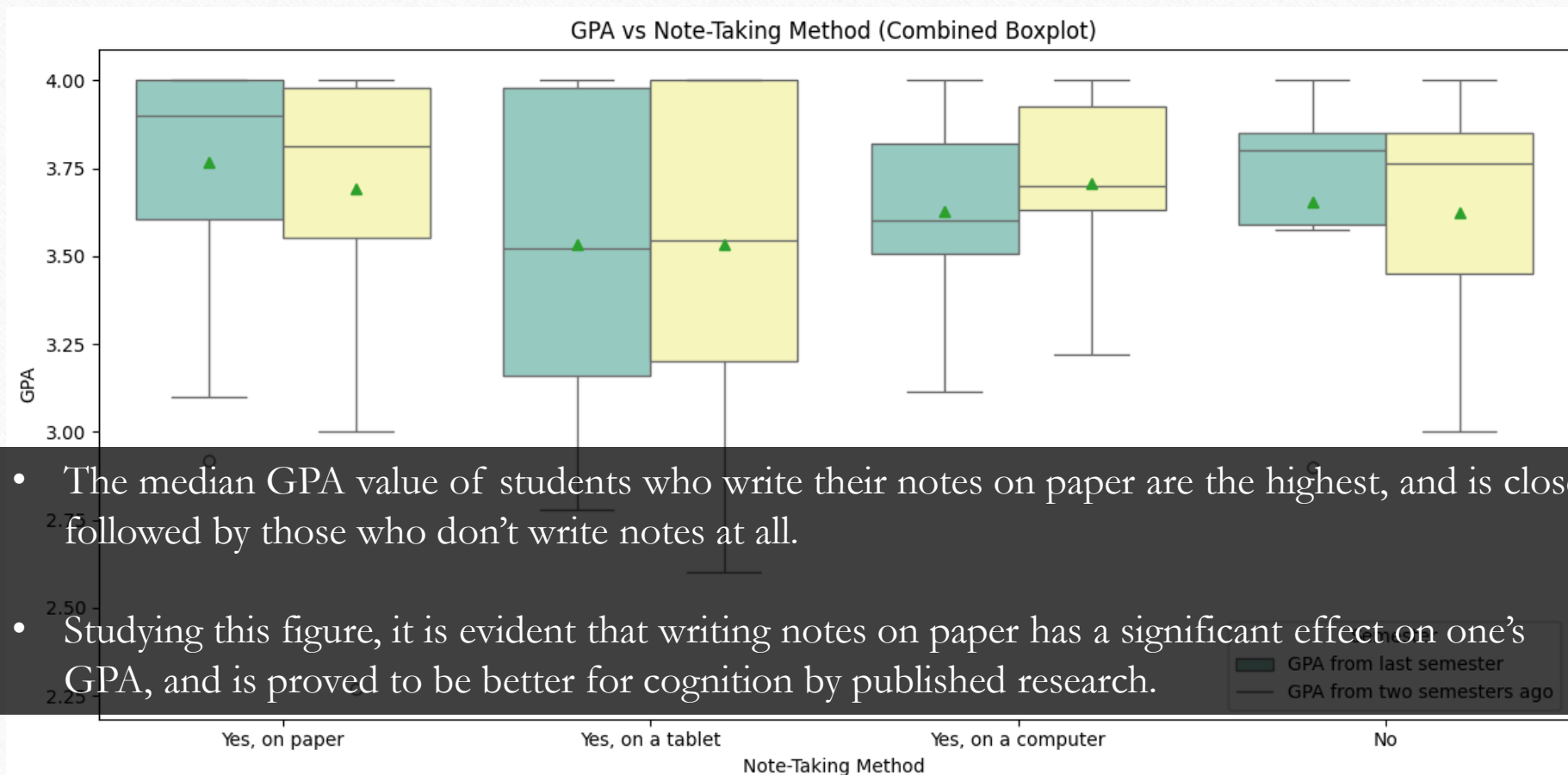
Does a private education influence the availability of a scholarship to students?

- Presuming the Null hypothesis that this may occur, a chi-squared test yields a statistically insignificant relationship regarding our hypothesis.
- A perfectly independent relationship is observed
 - chi-squared statistic of 0 representing no deviation
 - p-value of 1, proving the null hypothesis to be false.

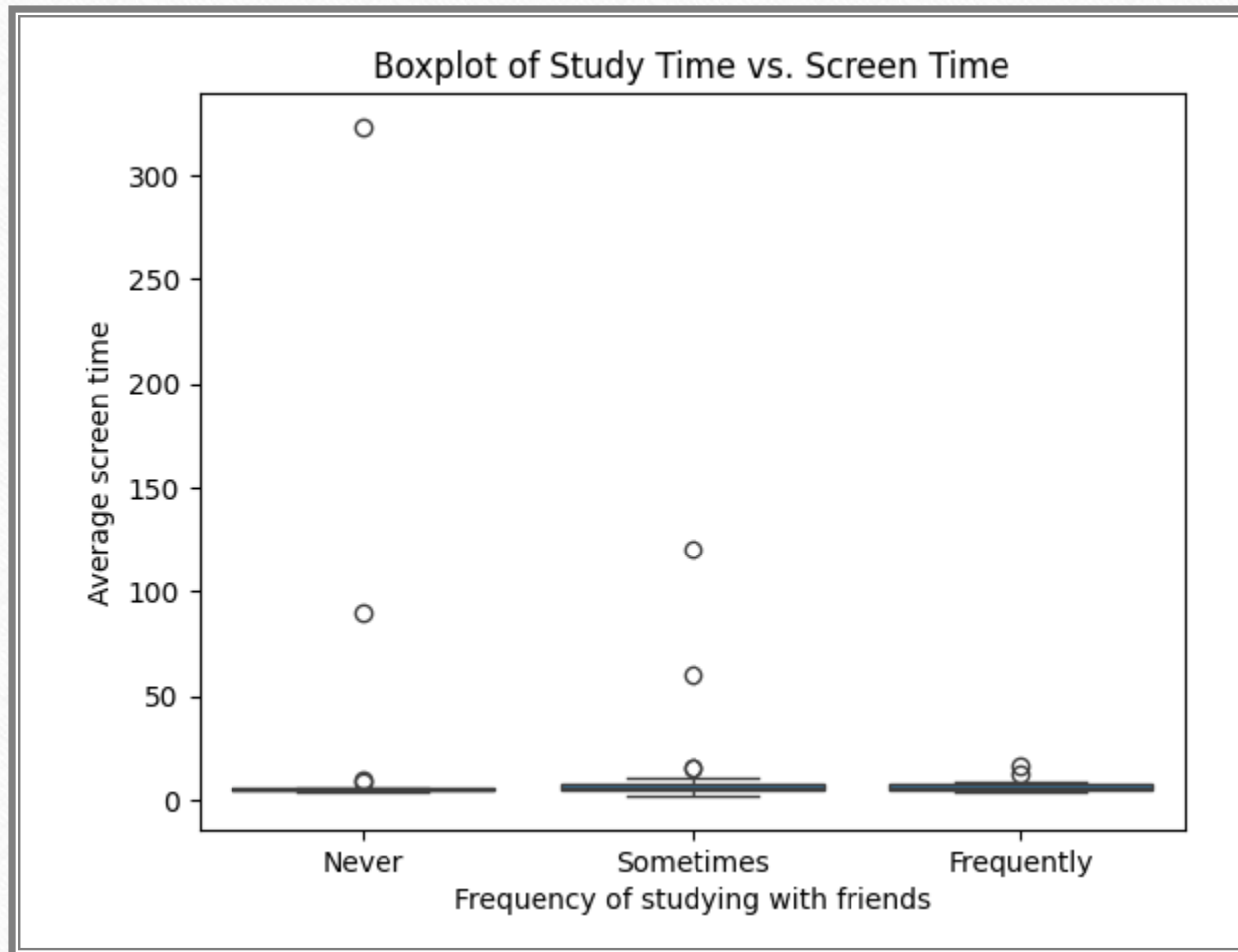
Does a particular method of notetaking correspond to a higher GPA?



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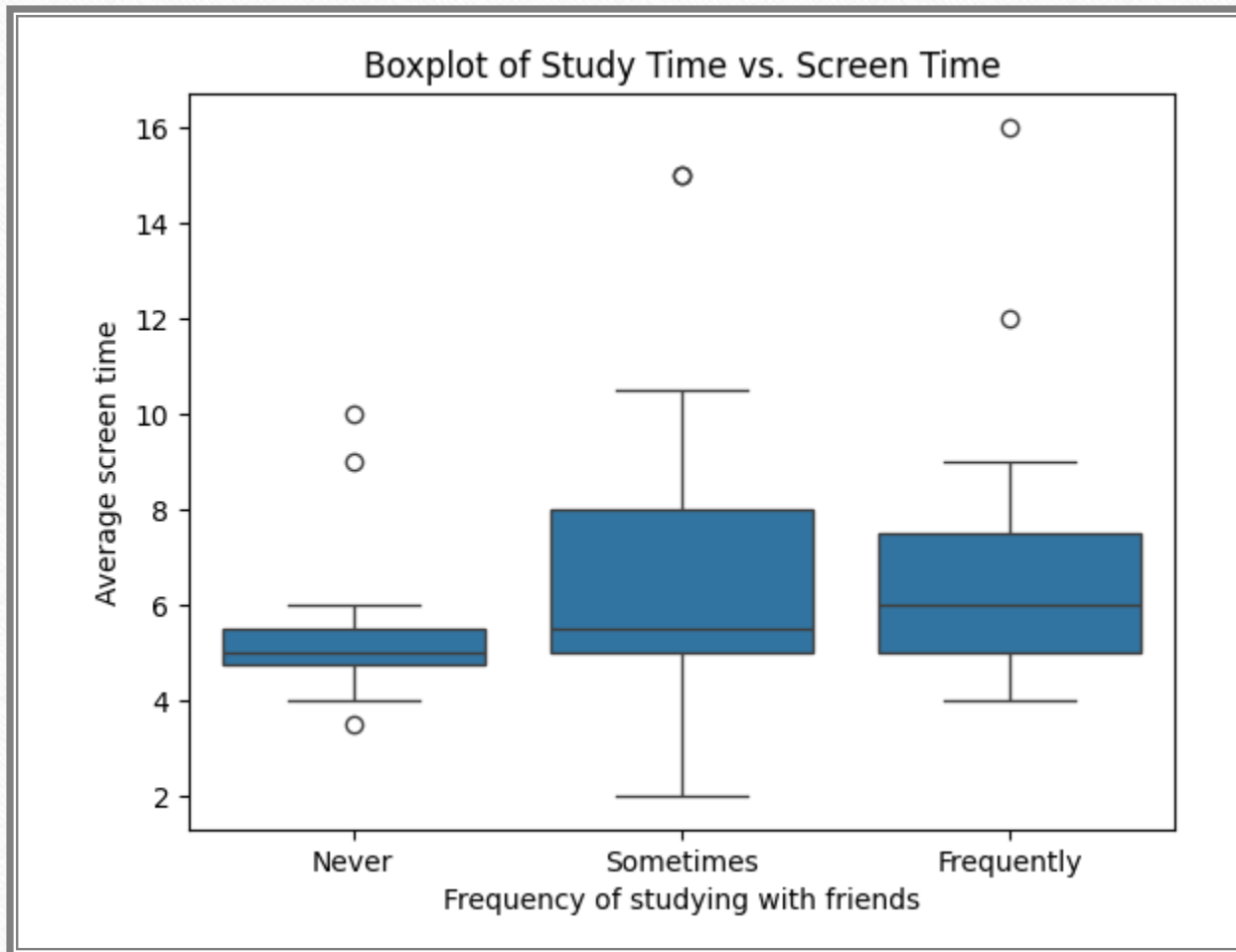


- The median GPA value of students who write their notes on paper are the highest, and is closely followed by those who don't write notes at all.
- Studying this figure, it is evident that writing notes on paper has a significant effect on one's GPA, and is proved to be better for cognition by published research.



Is there a correlation between students who study with friends and their average screen time?

- At first it looks peculiar, with multiple outliers influencing the data, rendering a Pearson's correlation coefficient of -0.18 suggesting there is no correlation.



Is there a correlation between students who study with friends and their average screen time?

- After disregarding the outliers in the data, the graph is easier to understand and reveals a more directly proportional relationship between the frequency of studying with friends and the students' average screen time.
- The Pearson's coefficient is now around 0.2, suggesting some correlation between the two.

Does having a parent in tech have an impact on the starting age for programming?

No it does not!

The Pearson's
coefficient correlation
gives a value of -0.28

Additionally, an
Ordinary Least Squares
(OLS) Regression model
marks this negative
relation as -1.16

Conclusion

- This presentation covered some general hypotheses regarding UMD's student response dataset, namely-
 - Does a private education influence the availability of a scholarship to students?
 - Does a particular method of notetaking correspond to a higher GPA?
 - Is there a correlation between students who study with friends and their average screen time?
 - Does having a parent in tech have an impact on the starting age for programming?
- Using chi-squared tests, graphical representations and correlation analysis, definitive answers were derived for each hypothesis.